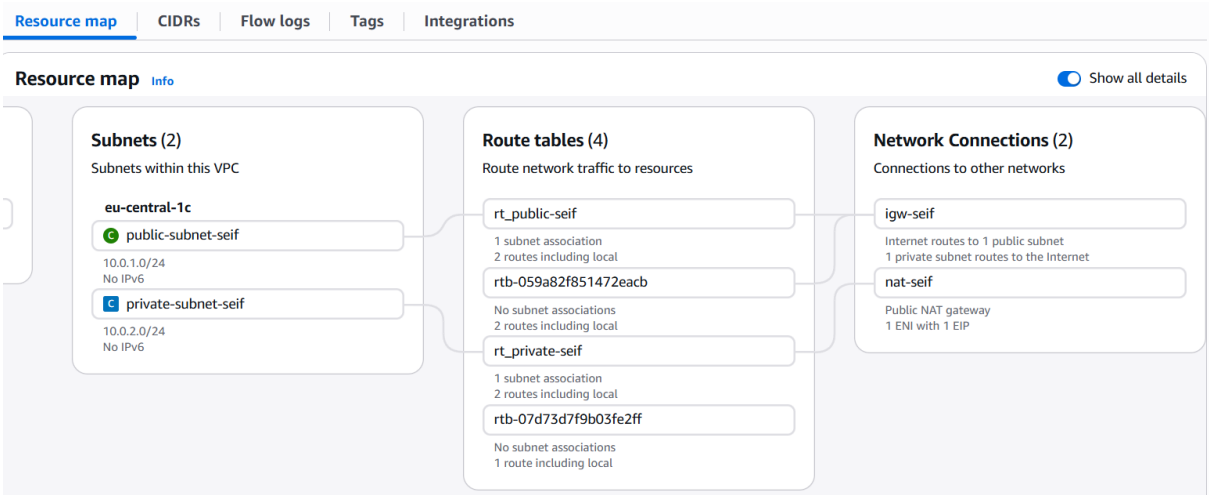
LAB – VPC



1. Why does the NAT Gateway need to be in the public subnet?
Because it needs internet access to send traffic out for the private subnet.

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2. What if you don't associate the public subnet with the public route table?
Resources in that subnet won't have internet access.

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3. Internet Gateway vs NAT Gateway

- Internet Gateway: allows inbound and outbound traffic
- NAT Gateway: allows outbound only (no inbound)

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1. **Can you SSH into a private EC2 from your laptop?**
   No as it has no direct internet access.

web-lab

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash - devops-labs

```
seife@SeifEddine MINGW64 ~/web-lab/web-lab (main)
$ cd

seife@SeifEddine MINGW64 ~
$ mv ~/Downloads/labseifvpc.pem ~/devops-labs/
cd ~/devops-labs

seife@SeifEddine MINGW64 ~/devops-labs (main)
$ chmod 400 labseifvpc.pem

seife@SeifEddine MINGW64 ~/devops-labs (main)
$ ssh -i labseifvpc.pem ec2-user@3.68.115.38
The authenticity of host '3.68.115.38 (3.68.115.38)' can't be established.
ED25519 key fingerprint is: SHA256:bOkcwF1zaQgZQzt7+MoxRFnAYk1ofJ4ZZ2UgKM1GHSQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.68.115.38' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
       #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
[ec2-user@ip-10-0-1-13 ~]$
```

aws Europe (Frankfur ▾ Account ID: 6866-9977-4... dvpt-saief

EC2 > Instances > i-06b89f39280cf674e

EC2

Dashboard
AWS Global View
Events
▾ Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
▾ Images
AMIs
AMI Catalog
▾ Elastic Block Store
Volumes

Instance ID
i-06b89f39280cf674e

Public IPv4 address
Private IPv4 address copied
10.0.1.13

IPv6 address
–

Instance state
Running

Public DNS
–

Hostname type
IP name: ip-10-0-1-13.eu-central-1.compute.internal

Private IP DNS name (IPv4 only)
ip-10-0-1-13.eu-central-1.compute.internal

Answer private resource DNS name
–

CloudShell Feedback Console Mobile App Privacy Terms Cookie preferences

```
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$ cd

seife@SeifEddine MINGW64 ~
$ mv ~/Downloads/labseifvpc.pem ~/devops-labs/
cd ~/devops-labs

seife@SeifEddine MINGW64 ~/devops-labs (main)
$ chmod 400 labseifvpc.pem

seife@SeifEddine MINGW64 ~/devops-labs (main)
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       #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
[ec2-user@ip-10-0-1-13 ~]$ ssh ec2-user@10.0.1.13
The authenticity of host '10.0.1.13 (10.0.1.13)' can't be established.
ED25519 key fingerprint is SHA256:bOkcwF1zaQgZQzt7+MoxRFnAYk1ofJ4ZZ2UgKM1GHSQ.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.1.13' (ED25519) to the list of known hosts.
ec2-user@10.0.1.13: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-0-1-13 ~]$
```

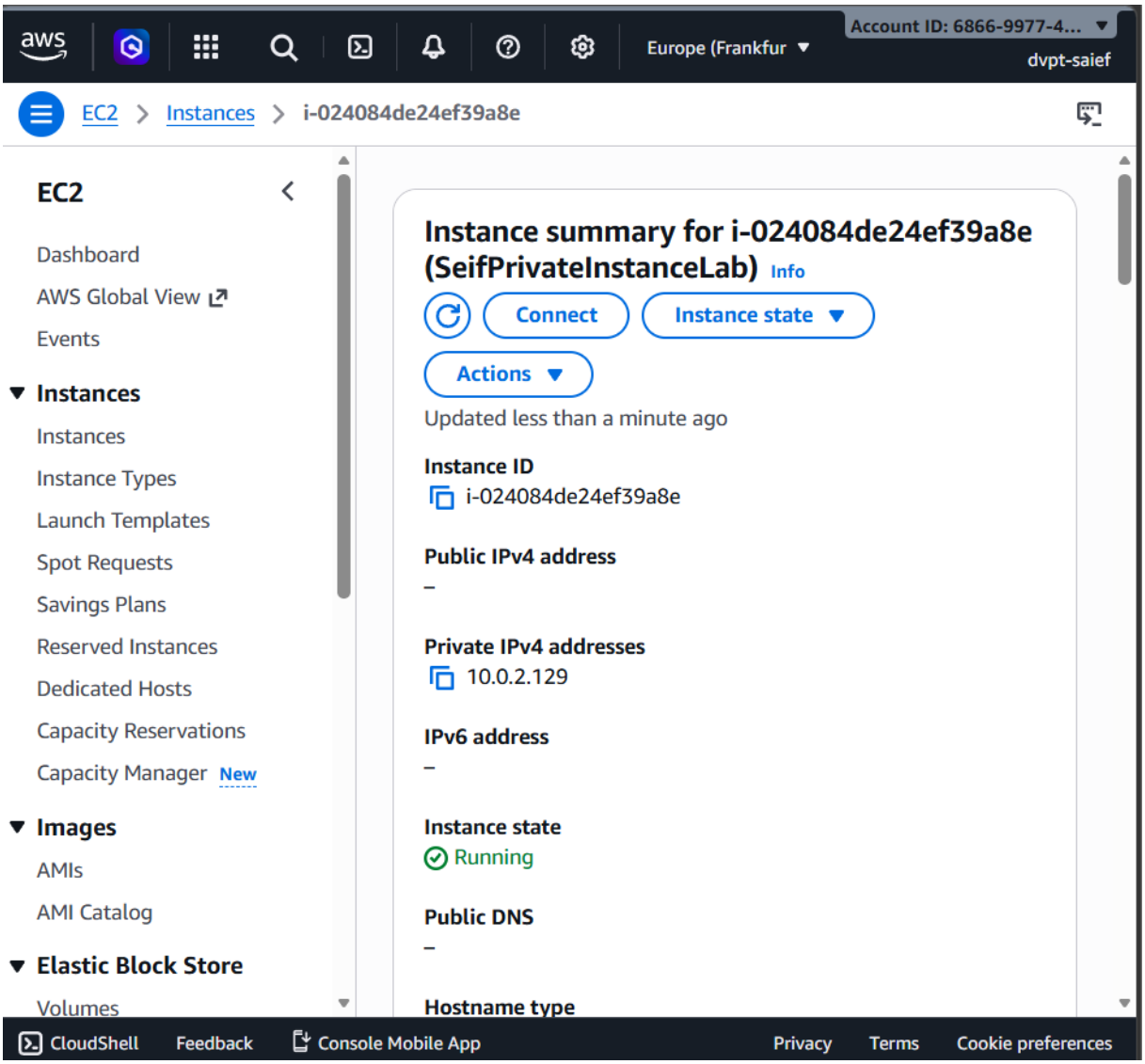
aws
Europe (Frankfur
Account ID: 6866-9977-4...
dvpt-saief
EC2 > Instances > i-024084de24ef39a8e
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instance summary for i-024084de24ef39a8e (SeifPrivateInstanceLab) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-024084de24ef39a8e
Public IPv4 address
–
Private IPv4 addresses
10.0.2.129
IPv6 address
–
Instance state
Running
Public DNS
–
Hostname type
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences

```
seife@SeifEddine MINGW64 ~/devops-labs (main)
$ ssh -i seiflabvpc.pem ec2-user@18.195.21.170
  ~~      \###|
  ~~        \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~        V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-1-30 ~]$ ssh -A -i labseifvpc.pem ec2-user@18.195.21.170
Warning: Identity file labseifvpc.pem not accessible: No such file or directory.
The authenticity of host '18.195.21.170 (18.195.21.170)' can't be established.
ED25519 key fingerprint is SHA256:RxzRpCmsGm1qgIpwgaxomZfBwlbCt6Gq11jUFk5MUsw.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.195.21.170' (ED25519) to the list of known hosts.
ec2-user@18.195.21.170: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-0-1-30 ~]$ ssh ec2-user@10.0.2.129
The authenticity of host '10.0.2.129 (10.0.2.129)' can't be established.
ED25519 key fingerprint is SHA256:NTG0eB7s/UG79q/ItB5lEPyDQgBx9AyI3X0VUM0Q0AA.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.129' (ED25519) to the list of known hosts.
ec2-user@10.0.2.129: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-0-1-30 ~]$ ping google.com
PING google.com (192.178.183.101) 56(84) bytes of data.
64 bytes from lcfraw-in-f101.1e100.net (192.178.183.101): icmp_seq=1 ttl=113 time=0.449
ms
64 bytes from lcfraw-in-f101.1e100.net (192.178.183.101): icmp_seq=2 ttl=113 time=0.537
ms
64 bytes from lcfraw-in-f101.1e100.net (192.178.183.101): icmp_seq=3 ttl=113 time=0.522
ms
64 bytes from lcfraw-in-f101.1e100.net (192.178.183.101): icmp_seq=4 ttl=113 time=0.532
ms
64 bytes from lcfraw-in-f101.1e100.net (192.178.183.101): icmp_seq=5 ttl=113 time=0.469
ms
64 bytes from lcfraw-in-f101.1e100.net (192.178.183.101): icmp_seq=6 ttl=113 time=0.531
ms
64 bytes from lcfraw-in-f101.1e100.net (192.178.183.101): icmp_seq=7 ttl=113 time=0.550
ms
```

```
seife@SeifEddine MINGW64 ~
$ ssh -A ec2-user@3.120.249.212
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.120.249.212' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Tue Apr  7 21:06:07 2026 from 93.31.5.9
[ec2-user@ip-10-0-1-30 ~]$ ssh ec2-user@10.0.2.240
The authenticity of host '10.0.2.240 (10.0.2.240)' can't be established.
ED25519 key fingerprint is SHA256:nDiMChbHop1UJevpmBkYgo5nHj39Qvg5xBOwSTdsFF4.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.240' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-2-240 ~]$ ping google.com
PING google.com (142.250.154.139) 56(84) bytes of data.
64 bytes from bt-in-f139.1e100.net (142.250.154.139): icmp_seq=1 ttl=113 time=1.30 ms
64 bytes from bt-in-f139.1e100.net (142.250.154.139): icmp_seq=2 ttl=113 time=0.921 ms
64 bytes from bt-in-f139.1e100.net (142.250.154.139): icmp_seq=3 ttl=113 time=0.917 ms
64 bytes from bt-in-f139.1e100.net (142.250.154.139): icmp_seq=4 ttl=113 time=0.929 ms
64 bytes from bt-in-f139.1e100.net (142.250.154.139): icmp_seq=5 ttl=113 time=0.886 ms
64 bytes from bt-in-f139.1e100.net (142.250.154.139): icmp_seq=6 ttl=113 time=0.910 ms
```

Private instance needed to be removed, and recreated due to some issues related to the key. Final screenshot has everything fixed, mainly due to a very minor error on my part regarding the subnet (instead of private, public was selected).

All done!